KENT PEARCE **MATH 1351 Spring 2004**

Homework Set HW8 due 3/22/04 at 12:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)
This is problem 1 Section 4.1 page 193.

Find the critical numbers for $f(x) = 5 + 10x - x^2$ on the interval $[-3, 3]$. Then decide whether the critical value is a maximum (MAX), minimum (MIN) or neither (N).

For this problem there are two critical numbers $x_1 < x_2$ in the given interval:

$x_1 =$ _____ (MAX, MIN, N) _____
$x_2 =$ _____ (MAX, MIN, N) _____

2.(1 pt)
This is problem 4 Section 4.1 page 193.

Find the critical numbers for $f(t) = t^4 - 8t^2$ on the interval $[-3, 3]$. Then decide whether the critical value is a maximum (MAX), minimum (MIN) or neither (N).

For this problem there are five critical numbers $x_1 < x_2 < x_3 < x_4 < x_5$ in the given interval:

$x_1 =$ _____ (MAX, MIN, N) _____
$x_2 =$ _____ (MAX, MIN, N) _____
$x_3 =$ _____ (MAX, MIN, N) _____
$x_4 =$ _____ (MAX, MIN, N) _____
$x_5 =$ _____ (MAX, MIN, N) _____

3.(1 pt)
This is problem 8 Section 4.1 page 193.

Find the critical numbers for $g(t) = 3t^5 - 20t^3$ on the interval $[-1, 2]$. Then decide whether the critical value is a maximum (MAX), minimum (MIN) or neither (N).

For this problem there are three critical numbers $t_1 < t_2 < t_3$ in the given interval:

$t_1 =$ _____ (MAX, MIN, N) _____
$t_2 =$ _____ (MAX, MIN, N) _____
$t_3 =$ _____ (MAX, MIN, N) _____

4.(1 pt)
This is problem 10 Section 4.1 page 193.

Find the critical numbers for $s(x) = \dfrac{\ln(\sqrt{x})}{x}$ on the interval $[1, 3]$. Then decide whether the critical value is a maximum (MAX), minimum (MIN) or neither (N).

For this problem there are three critical numbers $x_1 < x_2 < x_3$ in the given interval:

$x_1 =$ _____ (MAX, MIN, N) _____
$x_2 =$ _____ (MAX, MIN, N) _____
$x_3 =$ _____ (MAX, MIN, N) _____

5.(1 pt)
This is problem 12 Section 4.1 page 193.

Find the critical numbers for $f(x) = |x - 3|$ on the interval $[-4, 4]$. Then decide whether the critical value is a maximum (MAX), minimum (MIN) or neither (N).

For this problem there are three critical numbers $x_1 < x_2 < x_3$ in the given interval:

$x_1 =$ _____ (MAX, MIN, N) _____
$x_2 =$ _____ (MAX, MIN, N) _____
$x_3 =$ _____ (MAX, MIN, N) _____

6.(1 pt)
This is problem 20 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $g(x) = 2x^3 - 3x^2 - 36x + 4$ on the interval $[-4, 4]$.

Absolute Maximum: $x =$ _____ $g(x) =$ _____
Absolute Minimum: $x =$ _____ $g(x) =$ _____

7.(1 pt)
This is problem 22 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $f(x) = \dfrac{8}{3}x^3 - 5x^2 + 8x - 5$ on the interval $[-4, 4]$.

Absolute Maximum: $x =$ _____ $f(x) =$ _____
Absolute Minimum: $x =$ _____ $f(x) =$ _____

8.(1 pt)
This is problem 25 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $s(t) = t\cos(t) - \sin(t)$ on the interval $[0, 2\pi]$.

Absolute Maximum: $t =$ _____ $s(t) =$ _____
Absolute Minimum: $t =$ _____ $s(t) =$ _____

9.(1 pt)
This is problem 28 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $f(x) = \begin{cases} 9 - 4x & \text{for } x < 1 \\ -x^2 + 6x & \text{for } x \geq 1 \end{cases}$ on the interval $[0, 4]$.

NOTE: For this problem the maximum occurs at two x values, $x_1 < x_2$.

Absolute Maximum: $x_1 =$ _____, $x_2 =$ _____ $f(x) =$ _____
Absolute Minimum: $x =$ _____ $f(x) =$ _____

10.(1 pt)
This is problem 30 Section 4.1 page 194.

Find the smallest value of $f(x) = \dfrac{1}{x(x+1)}$ on the interval $[-1/2, 0]$.

If the smallest value on this interval does not exist then simply write *DNE* .

The smallest value is _____

11.(1 pt)
This is problem 32 Section 4.1 page 194.

Find the smallest value of $f(x) = \dfrac{(x^2 - 1)}{(x^2 + 1)}$ on the interval $[-1, 1]$.

If the smallest value on this interval does not exist then simply write *DNE* .

The smallest value is ____

12.(1 pt)
This is problem 36 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $f(\theta) = \cos^3(\theta) - 4\cos^2(\theta)$ on the interval $[-0.1, \pi + 0.1]$.

Absolute Maximum: $\theta =$ ____ $f(\theta) =$ ____

Absolute Minimum: $\theta =$ ____ $f(\theta) =$ ____

13.(1 pt)
This is problem 39 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $g(u) = 98u^3 - 4u^2 + 72u$ on the interval $[0,4]$.

Absolute Maximum: $u =$ ____ $g(u) =$ ____

Absolute Minimum: $u =$ ____ $g(u) =$ ____

14.(1 pt)
This is problem 40 Section 4.1 page 194.

Find the absolute maximum and absolute minimum of $f(w) = \sqrt{w}(w-5)^{1/3}$ on the interval $[0,4]$.

Absolute Maximum: $w =$ ____ $f(w) =$ ____

Absolute Minimum: $w =$ ____ $f(w) =$ ____

15.(1 pt)
This is problem 4 Section 4.2 page 199.

Verify that the function f satisfies the hypotheses of the MVT on the given interval $[a,b]$. Then find all numbers c between a and b for which

$$\frac{f(b)-f(a)}{b-a} = f'(c).$$

$$f(x) = -x^2 + 4, \quad \text{on } [-1,0]$$

To do this problem you will be asked to carry out some intermediate steps:

(1) Find $f'(x) =$ ____________

(2) Find $\dfrac{f(b)-f(a)}{b-a} =$ ____

(3) There is one value of c :

$c =$ ____

16.(1 pt)
This is problem 6 Section 4.2 page 199.

Verify that the function f satisfies the hypotheses of the MVT on the given interval $[a,b]$. Then find all numbers c between a and b for which

$$\frac{f(b)-f(a)}{b-a} = f'(c).$$

$$f(x) = 2x^3 - x^2, \quad \text{on } [0,2]$$

To do this problem you will be asked to carry out some intermediate steps:

(1) Find $f'(x) =$ ____________

(2) Find $\dfrac{f(b)-f(a)}{b-a} =$ ____

(3) There is one value of c :

$c =$ ____

17.(1 pt)
This is problem 26 Section 4.2 page 200.

Decide whether Rolle's Theorem can be applied to the function f on the given interval $[a,b]$.

$$f(x) = \frac{1}{x-2}, \quad \text{on } [-1,1]$$

For this problem, enter *YES* if Rolle's theorem is applicable and *NO* if it is not.

YES or NO : ____

18.(1 pt)
This is problem 28 Section 4.2 page 200.

Decide whether Rolle's Theorem can be applied to the function f on the given interval $[a,b]$.

$$f(x) = 3x + \sec(x), \quad \text{on } [-\pi, \pi]$$

For this problem, enter *YES* if Rolle's theorem is applicable and *NO* if it is not.

YES or NO : ____